

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

CAUTION

Use caution when disconnecting or removing fuel lines, replacing filters and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

CAUTION

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

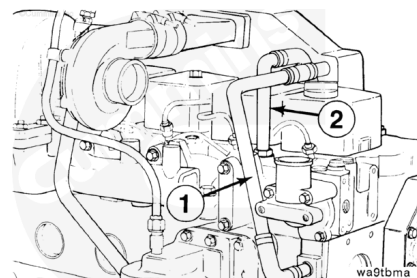
- Close the sea water supply and return line(s). See equipment manufacturer service information.
- Close the fuel supply and return lines. See equipment manufacturer service information.
- Disconnect the battery power from the engine. See equipment manufacturer service information.
- Drain the sea water by removing the zinc plug at the bottom of the aftercooler assembly. Refer to Procedure 008-059 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-059.html>)
- Remove the following, if equipped. Refer to Procedure 006-062 in Section 6. (</qs3/pubsys2/xml/en/procedures/40/40-006-062.html>)

- Fuel cooler lower molded sea water hose to the aftercooler
- Fuel cooler clip and mounting capscrew
- Fuel drain flex hoses at the fuel cooler.
- Loosen the aftercooler top and bottom air connection hose clamps. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html>)
- Remove the aftercooler air box drain plug and drain the condensation from the aftercooler housing core.
- Disconnect the aftercooler air inlet connection hose. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Loosen the hose clamp on the water transfer connection. Remove the aftercooler sea water discharge hose.
- Loosen the hose clamps connecting the aftercooler to the marine gear oil cooler. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-041.html>)
- Detach the oil dipstick tube if connected to the aftercooler. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>) For jacket water aftercoolers: Drain 2 liters [2.1 qt] of coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>)

Remove

Remove the coolant supply tube (1) and the coolant return tube (2).

Remove the aftercooler housing and gasket.



Install

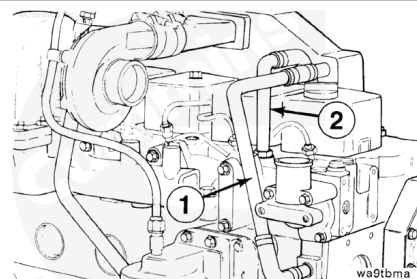
Note : The holes shown in the illustration are drilled through. Apply liquid Teflon™ sealant to the capscrews.

Install the aftercooler housing and a new gasket.

Torque Value: 24 n•m [212 in-lb]

Install the coolant supply tube (1) and coolant return tube (2).

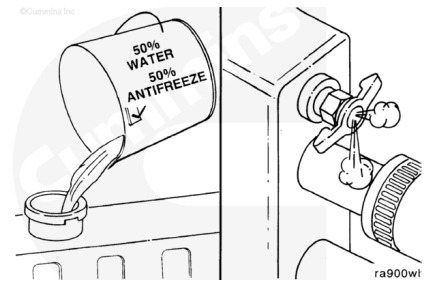
Torque Value: 8 n•m [71 in-lb]



Fill

⚠ CAUTION ⚠

During filling, air must be vented from the engine cooling passages. Open the engine vent petcock. Make sure to open the petcock on the aftercooler for aftercooled engines. The system must be filled slowly to prevent air locks. Wait 2 to 3 minutes to allow air to be vented; then add coolant to bring the level to the bottom of the radiator filler neck. Failure to do so will cause entrapment of air in cooling system and will cause engine to overheat.



Fill the coolant system with a premixture of 50 percent water and 50 percent ethylene-glycol-type antifreeze.

Finishing Steps

⚠ WARNING ⚠

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⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ CAUTION ⚠

Use caution when disconnecting or removing fuel lines, replacing filters and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

⚠ CAUTION ⚠

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

- Attach the oil dipstick tube, if connected, to the aftercooler. Refer to Procedure 007-011 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Install the following, if removed. Refer to Procedure 006-062 in Section 6. (</qs3/pubsys2/xml/en/procedures/40/40-006-062.html>)
 - Fuel cooler lower molded sea water hose to the aftercooler
 - Fuel cooler clip and mounting capscrew
 - Fuel drain flex hoses at the fuel cooler.
- Connect the aftercooler air inlet connection to the turbocharger. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html>)
- Tighten the aftercooler top and bottom air connection hose clamps.
- Install the dipstick tube brace. Refer to Procedure 007-011 in Section 11. (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Connect the battery power to the engine. See equipment manufacturer service information.
- Open the fuel supply and return lines. See equipment manufacturer service information.
- Operate the engine. Check for leaks.